Diet Planner Based on BMI

Project Report

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Abstract

This report presents a machine learning-based diet recommendation system that predicts daily caloric requirements using BMI, age, gender, medical conditions, and activity level. A Random Forest regression model is used, integrated into a Gradio web interface for interactive user access.

1 Introduction

Healthy diet planning requires personalization based on a person's physical metrics, medical history, and lifestyle. This project builds an AI-powered system that estimates caloric needs and offers tailored dietary suggestions.

2 Literature Review

Recent research demonstrates increasing interest in AI-driven nutrition support:

- **Golagana et al. (2023)** employed Random Forest, K-Means, and LSTM models for diet personalization [1].
- **Tsolakidis et al. (2024)** conducted a large-scale review of AI for nutrition across 67 studies [2].

2.1 Identified Gaps

1. Most diet planners are generic and ignore BMI or disease-specific needs.
2. Medical indicators (e.g., cholesterol) rarely influence recommendations.
3. Absence of accessible, automatic, AI-driven nutrition tools for users.

3 Dataset

The dataset includes:

- Weight, Height, Age
- Gender
- Disease Type (diabetes, hypertension, none)
- Activity Level (low/medium/high)

- Cholesterol (optional)
- Caloric Requirement (target)

4 Methodology

4.1 Data Processing

- Median imputation for numerical fields.
- Mode imputation + one-hot encoding for categorical fields.
- Standardization for numeric attributes.

4.2 Feature Engineering

BMI computed as:

$$\text{BMI} = \frac{\text{weight (kg)}}{(\text{height (m)})^2}$$

5 Model Development

5.1 Algorithm Selection

A Random Forest Regressor was used due to:

- High performance on non-linear data
- Robustness to noise
- Feature importance interpretability

5.2 Evaluation Metrics

- Mean Absolute Error (MAE)
- RMSE
- R^2

6 Results

Metric	Train
MAE	515.485
RMSE	603.327
R^2	-0.103

Table 1: Evaluation Metrics

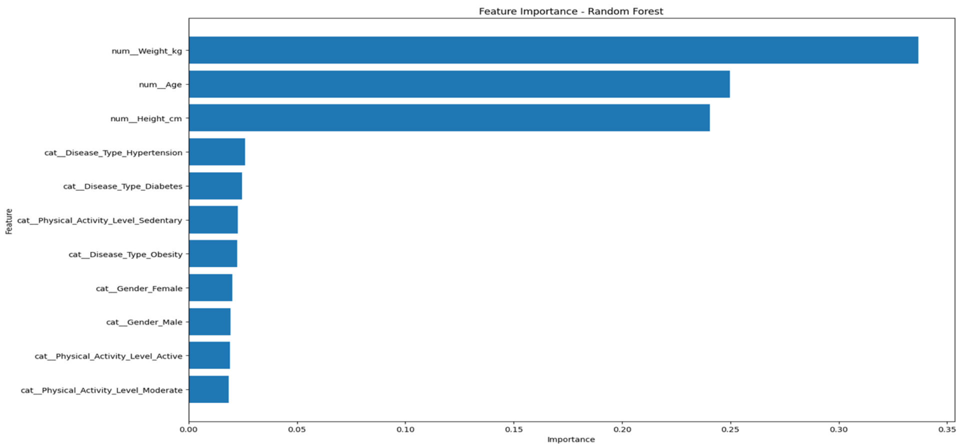

Figure 1: Feature importance chart

7 System Deployment

7.1 Web Interface

A Gradio interface displays:

- Predicted calorie requirement
- BMI
- Personalized diet suggestions

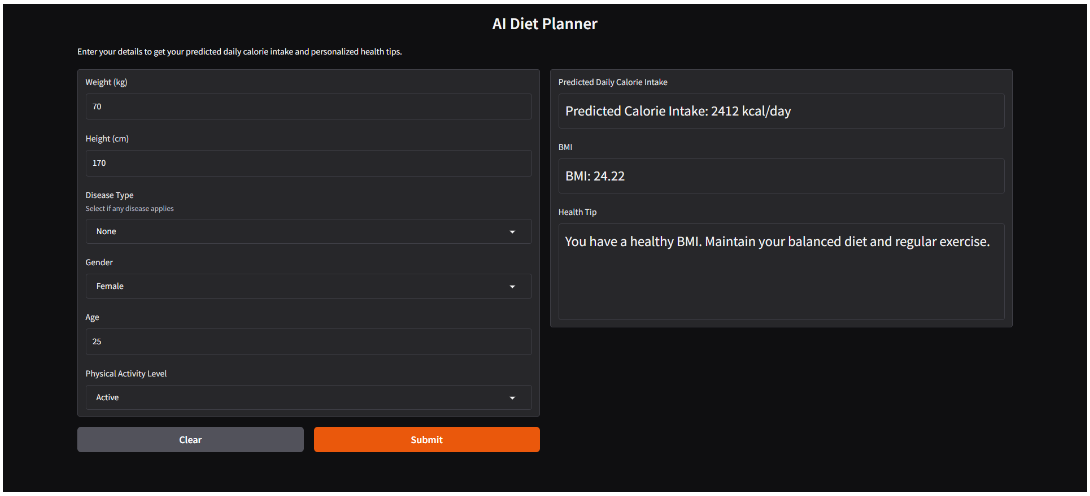

Figure 2: Gradio UI

8 Discussion

- Strengths: Lightweight, interpretable, easy to deploy.
- Limitations: Dataset size, medical validation required.
- Future Extensions: Macro-level diet planning, meal suggestions, deep learning models.

9 Conclusion

This project successfully demonstrates an AI-powered diet planner combining ML predictions with an interactive interface.

References

[1] R. Golagana, V. Sravani, and T. M. Reddy, "Diet recommendation system using machine learning," *Unpublished / Conference Paper*, 2023, Described use of Random Forest, K-Means and LSTM for personalized diet plans.

[2] D. Tsolakidis, L. P. Gymnopoulos, and K. Dimitropoulos, "Artificial intelligence and machine learning technologies for personalized nutrition: A review," *Survey / Review*, 2024, Review of AI/ML for personalized nutrition (67 studies, 2021–2024).

[3] M. Patel and K. Karia, *Diet planner based on bmi – project report*, Project artifacts and slides: see attached PPTX, 2025.